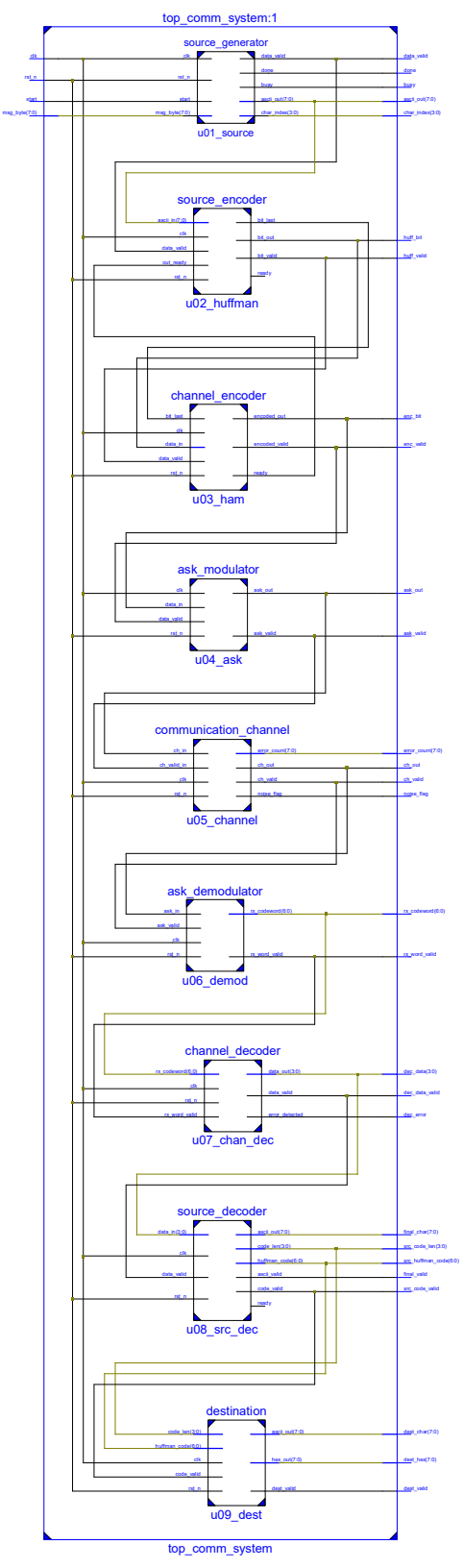

top_comm_system:1
source_generator
clk
rst_n
start
msg_byte[7:0]
data_valid
done
busy
ascii_out[7:0]
char_index[3:0]
u01_source
source_encoder
ascii_in[7:0]
clk
data_valid
out_ready
rst_n
bit_last
bit_out
bit_valid
ready
u02_huffman
channel_encoder
bit_last
clk
data_in
data_valid
rst_n
encoded_out
encoded_valid
ready
u03_ham
ask_modulator
clk
data_in
data_valid
rst_n
ask_out
ask_valid
u04_ask
communication_channel
ch_in
ch_valid_in
clk
rst_n
error_count[7:0]
ch_out
ch_valid
noise_flag
u05_channel
ask_demodulator
ask_in
ask_valid
clk
rst_n
rx_codeword[6:0]
rx_word_valid
u06_demod
channel_decoder
rx_codeword[6:0]
clk
rst_n
rx_word_valid
data_out[3:0]
data_valid
error_detected
u07_chan_dec
source_decoder
data_in[3:0]
clk
data_valid
rst_n
ascii_out[7:0]
code_len[3:0]
huffman_code[6:0]
ascii_valid
code_valid
ready
u08_src_dec
destination
code_len[3:0]
huffman_code[6:0]
clk
code_valid
rst_n
ascii_out[7:0]
hex_out[7:0]
dest_valid
u09_dest
data_valid
done
busy
ascii_out[7:0]
char_index[3:0]
huff_bit
huff_valid
enc_bit
enc_valid
ask_out
ask_valid
error_count[7:0]
ch_out
ch_valid
noise_flag
rx_codeword[6:0]
rx_word_valid
dec_data[3:0]
dec_data_valid
dec_error
final_char[7:0]
src_code_len[3:0]
src_huffman_code[6:0]
final_valid
src_code_valid
dest_char[7:0]
dest_hex[7:0]
dest_valid
top_comm_system